

Unnamed_Unit

1. Real numbers

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Real Numbers

Real number is any number that can be found in the real world. We find numbers everywhere around us. Natural numbers are used for counting objects, rational numbers are used for representing fractions, irrational numbers are used for calculating the square root of a number, integers for measuring temperature, and so on. These different types of numbers make a collection of real numbers. In this lesson, let us learn all about **what are real numbers, the subsets of real numbers** along with real numbers examples.

What are Real Numbers?

Any number that we can think of, except complex numbers, is a real number. For example, 3, 0, 1.5, $\frac{3}{2}$, $\sqrt{5}$, and so on are real numbers.

Definition of Real Numbers

Real numbers include rational numbers like positive and negative integers, fractions, and irrational numbers. Now, which numbers are not real numbers? The numbers that are neither rational nor irrational are non-real numbers, like, $\sqrt{-1}$, $2 + 3i$, and $-i$. These numbers include the set of [complex numbers](#), C .

Set of Real Numbers

The set of real numbers, which is denoted by R , is the union of the set of rational numbers (Q) and the set of irrational numbers ($\mathbb{Q}^c$). So, we can write the set of real numbers as, $R = Q \cup \mathbb{Q}^c$. This indicates that real numbers include natural numbers, whole numbers, integers, rational numbers, and irrational numbers. Observe the following table to understand this better. The table shows the [sets](#) of numbers that come under real numbers.

List of Real Numbers

The list of real numbers is endless because it includes all kinds of numbers like whole, natural, integers, rational, and irrational numbers. Since it includes integers it has negative numbers too. So, there is no specific number from which the list of real numbers starts or ends. It goes to infinity towards both sides of the number line.

Types of Real Numbers

We know that real numbers include rational numbers and irrational numbers. Thus, there does not exist any real number that is neither rational nor irrational. It simply means that if we pick up any number from R , it is either rational or irrational.

Rational Numbers

Any number which can be defined in the form of a fraction $\frac{p}{q}$ is called a rational number. The numerator in the fraction is represented as ' p ' and the denominator as ' q ', where ' q ' is not equal to zero. A rational number can be a natural number, a whole number, a decimal, or an integer. For example, $\frac{1}{2}$, $-\frac{2}{3}$, 0.5, 0.333 are rational numbers.

Irrational Numbers

Irrational numbers are the set of real numbers that cannot be expressed in the form of a fraction $\frac{p}{q}$ where ' p ' and ' q ' are integers and the [denominator](#) ' q ' is not equal to zero ($q \neq 0$). For example, π (π)

is an irrational number. $\pi = 3.14159265\dots$ In this case, the decimal value never ends at any point. Therefore, numbers like $\sqrt{2}$, $-\sqrt{7}$, and so on are irrational numbers.

Symbol of Real Numbers

Real numbers are represented by the symbol **R**. Here is a list of the symbols of the other types of numbers.

- **N** - Natural numbers
- **W** - Whole numbers
- **Z** - Integers
- **Q** - Rational numbers
- **Q'** - Irrational numbers

Subsets of Real Numbers

All numbers except complex numbers are real numbers. Therefore, real numbers have the following five subsets:

- **Natural numbers:** All positive counting numbers make the set of natural numbers, $N = \{1, 2, 3, \dots\}$
- **Whole numbers:** The set of natural numbers along with 0 represents the set of whole numbers. $W = \{0, 1, 2, 3, \dots\}$
- **Integers:** All positive counting numbers, negative numbers, and zero make up the set of integers. $Z = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- **Rational numbers:** Numbers that can be written in the form of a fraction p/q , where 'p' and 'q' are integers and 'q' is not equal to zero are rational numbers. $Q = \{-3, 0, -6, 5/6, 3.23\}$
- **Irrational numbers:** The numbers that are square roots of positive rational numbers, cube roots of rational numbers, etc., such as $\sqrt{2}$, come under the set of irrational numbers. $(\text{---}QQ\text{---}) = \{\sqrt{2}, -\sqrt{6}\}$

Real Numbers Chart

Among the sets given above, the sets N , W , and Z are the subsets of Q . The following figure shows the real numbers chart that explains the relationship between all the numbers mentioned above.

Properties of Real Numbers

Just like the set of natural numbers and integers, the set of real numbers also satisfies the closure property, the associative property, the commutative property, and the distributive property. The important properties of real numbers are mentioned below.

- **Closure Property:** The closure property states that the sum and product of two real numbers is always a real number. The closure property of $\mathbb{R}$ is stated as follows: If $a, b \in \mathbb{R}$, $a + b \in \mathbb{R}$ and $ab \in \mathbb{R}$
- **Associative Property:** The sum or product of any three real numbers remains the same even when the grouping of numbers is changed. The associative property of $\mathbb{R}$ is stated as follows: If $a, b, c \in \mathbb{R}$, $a + (b + c) = (a + b) + c$ and $a \times (b \times c) = (a \times b) \times c$

- **Commutative Property:** The sum and the product of two real numbers remain the same even after interchanging the order of the numbers. The commutative property of R is stated as follows: If $a, b \in R$, $a + b = b + a$ and $a \times b = b \times a$
- **Distributive Property:** Real numbers satisfy the distributive property. The distributive property of [multiplication](#) over [addition](#) is, $a \times (b + c) = (a \times b) + (a \times c)$ and the distributive property of multiplication over [subtraction](#) is $a \times (b - c) = (a \times b) - (a \times c)$

Real Numbers on a Number Line

A number line helps us to display numbers by representing them by a unique point on the line. Every point on the number line shows a unique real number. Note the following steps to represent real numbers on a number line:

- **Step 1:** Draw a horizontal line with arrows on both ends and mark the number 0 in the middle. The number 0 is called the origin.
- **Step 2:** Mark an equal length on both sides of the origin and label it with a definite scale.
- **Step 3:** It should be noted that the positive numbers lie on the right side of the origin and the negative numbers lie on the left side of the origin.

Observe the numbers highlighted on the number line. It shows real numbers like $-5/2$, 0, $3/2$, and 2.

Difference Between Real Numbers and Integers

The main difference between real numbers and integers is that real numbers include integers. In other words, integers come under the category of real numbers. Let us understand the difference between real numbers and integers with the help of the following table.